



## *M.Sc. Computer Science*

|                 |                                                 |
|-----------------|-------------------------------------------------|
| Division        | Faculty of Engineering and Technology           |
| School Name     | School of Computer Science & Engineering        |
| Department Name | Department of Computer Science and Applications |
| Programme Name  | M.Sc. Computer Science                          |

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## COURSE BASKET

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| Course Type          | Description                                                                                                    |
|----------------------|----------------------------------------------------------------------------------------------------------------|
| Programme Core       | Courses dealing with foundations, depth and breadth of the major in which student is admitted at MIT-WPU       |
| Programme Electives  | Open electives under the programme allow students to specialise in a particular area connected to their major. |
| University Core      | Courses that reflect the core MIT-WPU values and the mission of Life Transformation of students.               |
| University Electives | Multidisciplinary courses across the faculties at MIT-WPU and outside the programme core.                      |

### Semester I

| Semester | Course Type | Course Name / Course Title                                    | Total Credits |
|----------|-------------|---------------------------------------------------------------|---------------|
| I        | PM          | Advanced Java                                                 | 5             |
| I        | PM          | Advanced Operating System                                     | 4             |
| I        | PM          | Network Security                                              | 4             |
| I        | UC          | Research Methodology                                          | 4             |
| I        | PR          | Mini Project I                                                | 2             |
| I        | UC          | Scientific Studies of Mind, Matter, Spirit, and Consciousness | 2             |
| I        | UC          | Yoga                                                          | 1             |
|          |             | <b>Total:</b>                                                 | <b>22</b>     |

### Semester II

| Semester | Course Type | Course Name / Course Title         | Total Credits |
|----------|-------------|------------------------------------|---------------|
| II       | PM          | Algorithm Design Strategies        | 4             |
| II       | PM          | Python Programming                 | 4             |
| II       | PR          | Mini Project II                    | 2             |
| II       | PE          | Program Elective I (PE I)          | 4             |
| II       | PE          | Program Elective II (PE II)        | 4             |
| II       | PR          | Research Paper Writing             | 1             |
| II       | PR          | Cognitive Skills                   | 1             |
| II       | UC          | Peace Building: Global Initiatives | 2             |
| II       |             | <b>Total Credits:</b>              | <b>22</b>     |

### Semester - III

| Semester | Course Type | Course Name / Course Title    | Total Credits |
|----------|-------------|-------------------------------|---------------|
| III      | PM          | Full Stack Development        | 3             |
| III      | PM          | Cloud Computing               | 3             |
| III      | PM          | Machine Learning              | 4             |
| III      | PR          | Mini Project III              | 2             |
| III      | PR          | Lab on Full Stack Development | 1             |
| III      | PR          | Lab on Cloud Computing        | 1             |
| III      | PE          | Program Elective III (PE III) | 4             |
| III      | PE          | Program Elective IV (PE IV)   | 4             |
|          |             | <b>Total</b>                  | <b>22</b>     |

### SEMESTER - IV

| Semester | Course Type | Course Name / Course Title      | Total Credits |
|----------|-------------|---------------------------------|---------------|
| IV       | PR          | Full-time Industrial Internship | 22            |
|          |             | <b>Total Credits:</b>           | <b>22</b>     |

### Program Elective Tracks

| Semester | Course Type | Course Name / Course Title             |
|----------|-------------|----------------------------------------|
| II       | PE I        | Introduction to Cyber Security         |
| II       | PE I        | Data Warehousing and Data Mining       |
| II       | PE I        | Node JS                                |
| II       | PE I        | Software Testing and Quality Assurance |
| III      | PE II       | Cryptography                           |
| III      | PE II       | Business Intelligence                  |
| III      | PE II       | DevOps                                 |
| III      | PE II       | Information System Audit               |
| III      | PE II       | Digital Forensics                      |

\*Modifications to the programmes and courses are contingent upon adherence to university guidelines and procedures. Any proposed changes must undergo a thorough review process, including consultation with relevant academic departments, approval from the appropriate administrative bodies, and compliance with accreditation standards.

Additionally, consideration will be given to feedback from students, faculty, and other stakeholders to ensure that modifications align with the overall educational objectives and mission of the university. The implementation of any approved changes will be communicated transparently to the university community, and appropriate measures will be taken to facilitate a smooth transition for all affected parties.